Uncertainty-Guided Semisupervised Change Detection for Remote Sensing Images

Yupeng Gao, Tianyu Li, Pengxian Yin, Yiwen Hu, Shu Chen, and Wei Zhang

***Abstract*—Semisupervised change detection (CD) in remote sensing images faces a challenge wherein existing methods employ uniform processing strategies that fail to account for spatial heterogeneity in prediction difficulty, resulting in insufficient learning of difficult regions such as change boundaries and subtle variations that exhibit high uncertainty yet contain crucial discriminative information for accurate detection. To systematically address this limitation, this article proposes uncertainty-guided semisupervised CD (USCD), a novel framework that introduces a triple-guarantee mechanism for targeted learning reinforcement in difficult regions based on systematic uncertainty quantification. The framework comprises three synergistic components: uncertainty-aware protective augmentation (UAPA) employs window-based difficulty assessment to identify high-uncertainty regions and implements selective copy-paste augmentation that preserves critical information while maintaining data diversity; difficult region contrastive learning (DRCL) enhances feature discriminability in challenging areas through both local contrastive learning with anchor-based hard sample mining and global contrastive learning using prototype-based memory banks; and uncertainty-guided loss reweighting (UGLR) applies exponential adaptive weighting that strengthens supervision signals in high-uncertainty labeled regions while simultaneously suppressing noise propagation from unreliable pseudolabels in unlabeled data. Extensive experimental validation across six diverse benchmark datasets, including LEVIR-CD, WHU-CD, CDD, S2Looking, SYSU-CD, and JL1-CD, demonstrates that USCD achieves state-of-the-art performance with limited labeled data, showing substantial improvements, particularly concentrated in high-uncertainty regions, robust generalization capability validated through cross-domain transfer experiments, and competitive computational efficiency comparable to existing methods. Code is publicly available at https://github.com/G124556/USCD**

***Index Terms*—Change detection (CD), contrastive learning, pseudolabeling, semisupervised learning, uncertainty quantification.**

I. Introduction

THE field of change detection (CD) has witnessed remarkable progress through the integration of deep learning techniques, enabling automated analysis of multitemporal imagery [1], [2]. However, the field faces a persistent challenge that limits its practical impact: the enormous annotation effort required to train effective models [3], [4]. This challenge has motivated extensive research into semisupervised learning approaches that can leverage unlabeled data to reduce annotation requirements [5].

Current semisupervised CD methods predominantly rely on consistency regularization and pseudolabeling within teacher–student architectures [6], [7], [8]. These approaches enforce prediction consistency under different data augmentations or network perturbations, enabling models to learn from unlabeled data through self-supervision [9], [10]. While this paradigm has yielded improvements over supervised baselines given limited training data, existing methods overlook the spatial heterogeneity in learning difficulty across different regions, treating all spatial locations uniformly during the learning process.

The limitations can be validated through empirical observations in Fig. 1. In typical samples from the WHU-CD [11] dataset, we employ the uncertainty measure $U = 1 - |P_0 - P_1|$ (where P_0 and P_1 represents the prediction probabilities for unchanged and changed classes, respectively) to quantify the reliability of model predictions. The uncertainty map clearly reveals the spatially heterogeneous distribution of learning difficulty: high-uncertainty regions are primarily concentrated in two critical types: 1) boundary transition zones (red arrows), where gradual changes between classes create ambiguous predictions. This phenomenon is particularly pronounced in remote sensing images due to the pervasive mixed-pixel effect at object boundaries: a single pixel's spectral information simultaneously blends signals from both changed and unchanged land covers, which physically determines that P_0 and P_1 naturally converge at these locations, inevitably driving uncertainty U to elevated values. The high uncertainty at boundaries, therefore, does not reflect model deficiency, but rather constitutes a direct manifestation of remote sensing imaging characteristics in the prediction space; and 2) semantically confusing regions (blue arrows), where remote sensing imagery's top-down perspective induces genuine visual ambiguity across different land covers. For instance, large vehicles or infrastructure share highly similar texture and shape features with building rooftops under nadir viewing geometry, causing the model's predictions to persistently oscillate between the two classes and resulting in elevated uncertainty. Like boundary regions, the learning

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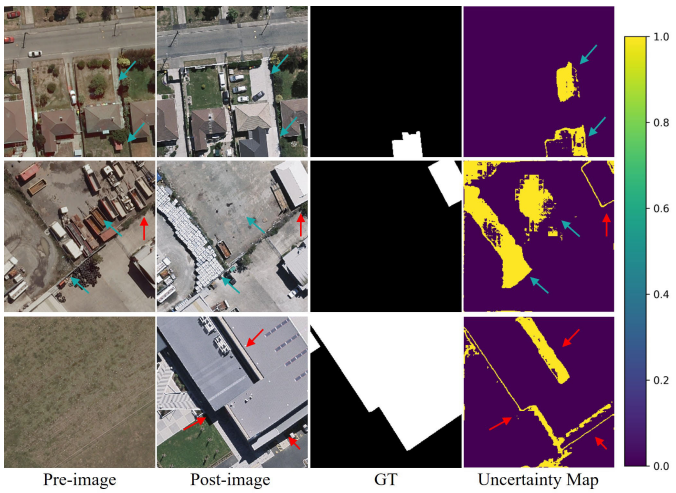


Fig. 1. Spatial distribution of prediction uncertainty in CD. High-uncertainty concentrates at change boundaries (red arrows) and confusing pseudochange regions (blue arrows). The uncertainty map reveals spatial heterogeneity that motivates our uncertainty-guided approach. Example from the WHU-CD dataset.

difficulty in these areas is rooted in the remote sensing imaging mechanism itself rather than random noise. The above analysis demonstrates that spatial heterogeneity in remote sensing CD is mechanistically determined to manifest as high uncertainty in the model prediction space. Uncertainty, therefore, serves not as an arbitrarily chosen surrogate, but as the natural projection of spatial heterogeneity onto the prediction space —representing an intrinsically principled measure of learning difficulty in CD. This insight directly motivates the design of our uncertainty-guided semisupervised CD (USCD) framework, in which uncertainty serves as the central organizing principle coordinating augmentation, contrastive learning, and adaptive supervision.

To operationalize this insight, we introduce USCD, which comprises three synergistic components designed to leverage uncertainty information. First, the uncertainty-aware protective augmentation (UAPA) module dynamically identifies high-uncertainty regions containing crucial discriminative information and protects them during data augmentation processes. Second, the difficult region contrastive learning (DRCL) mechanism designs specialized contrastive learning strategies specifically for difficult regions, strengthening the feature discriminability of challenging areas through hard sample mining and dynamic memory bank mechanisms. Finally, the uncertainty-guided loss reweighting (UGLR) strategy employs adaptive weight allocation, assigning greater weights to high-uncertainty regions in labeled data while assigning smaller weights to high-uncertainty regions in unlabeled data to suppress noise propagation. The main contributions can be summarized as follows.

1) An uncertainty-guided paradigm for semisupervised CD is introduced, where prediction uncertainty serves as the primary signal for identifying difficult regions and enabling differentiated learning. This formulation provides a new perspective on leveraging uncertainty within semisupervised CD.
2) An uncertainty-guided learning framework is developed, comprising three key components: UAPA, DRCL, and adaptive loss reweighting, enabling accurate identification and enhanced learning of difficult regions.
3) The proposed method is extensively evaluated on multiple standard datasets, showing clear performance improvements and strong cross-domain generalization, offering a reliable foundation for future research in semisupervised CD.

II. Related Work

A. Supervised CD

CD aims to identify land-cover changes between bitemporal remote sensing image pairs, and has become a fundamental task in Earth observation. Early deep learning methods based on fully convolutional networks and Siamese architectures established strong baselines for pixelwise change classification [12], [13]. Subsequently, Transformer-based methods such as BIT [14] and ChangeFormer [15] significantly improved the modeling of long-range dependencies between bitemporal features. More recent works further push detection accuracy through specialized architectures: Zhang et al. [16] propose a multiscale multiattention Siamese Transformer for capturing change patterns at multiple scales; MDGF-CD [17] embeds change-aware information into a DiffFormer backbone with multilevel feature grouping fusion for precise localization of subtle changes; and CAST [18] addresses hyperspectral scenarios with a cascade spectral-aware Transformer. Despite their strong performance, these fully supervised methods require extensive pixel-level annotations, which are costly to obtain in practice.

To alleviate annotation burdens, recent works explore data-level and supervision-level alternatives. Sample augmentation strategies have shown that data-level approaches can complement model-level advances under limited annotations [19], [20]. On the weakly supervised front, SWCD [21] introduces a similarity-aware strategy that reduces annotation requirements while maintaining competitive accuracy. Meanwhile, large-scale pretrained foundation models offer a new paradigm: HyperSIGMA [22] proposes a billion-parameter vision Transformer for hyperspectral interpretation with strong cross-task generalization and SkySense++ [23], [24] presents a multimodal foundation model pretrained on 27 million remote sensing images with impressive few-shot capabilities. Although these efforts have alleviated the annotation burden from different perspectives, they either still depend on carefully designed supervision signals or mainly improve data diversity and representation transfer without fully exploiting the abundant unlabeled bitemporal images available in practical applications.

B. Semisupervised CD

Semisupervised CD, as an important branch of remote sensing image analysis, aims to utilize limited labeled data and abundant unlabeled data for training to reduce annotation costs in practical applications [25], [26]. According to different learning strategies, existing methods can be mainly categorized

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into three types: self-training methods assign pseudolabels to unlabeled samples to enable supervised learning, but face challenges with label noise and error accumulation [27], [28]; adversarial learning methods employ discriminators to align prediction and label distributions, achieving effective data representation learning but often suffering from training instability [29], [30]; consistency learning methods enforce prediction consistency under different perturbations, providing stable supervision signals through consistency regularization [31], [32].

While these strategies relieve data scarcity from different perspectives, they commonly overlook a key property of CD imagery—the spatially heterogeneous distribution of learning difficulty. The majority of current approaches treat all spatial regions uniformly during training [33], [34], applying consistent processing strategies regardless of regional complexity. Given limited labeled data, this uniform treatment results in insufficient learning of difficult regions such as change boundaries and semantically ambiguous areas, where objects with similar visual characteristics to buildings (e.g., large vehicles, infrastructure) are prone to misclassification [35].

While a few recent works have begun to explore difficulty-aware strategies [36], they remain limited in two aspects. First, their difficulty identification lacks precision and generalization. These methods typically rely on simple heuristics such as entropy measures, which struggle to reliably distinguish between genuinely difficult samples containing critical discriminative information and unreliable pseudolabels caused by model errors or noise [37]. Second, even when difficult regions are identified, existing approaches lack systematic learning mechanisms to enhance feature discriminability in these areas. Most methods merely apply coarse adjustments such as confidence-based loss reweighting or sample rebalancing [38], [39]. These limitations motivate us to develop an uncertainty-guided framework that accurately quantifies spatially heterogeneous learning difficulty and designs targeted learning strategies to enhance feature learning in difficult regions while maintaining high-quality pseudolabels.

C. Uncertainty-Guided Learning

Uncertainty quantification has proven effective in improving model performance across various computer vision tasks [35], [40], [41], [42]. In semantic segmentation, uncertainty-guided sample selection enables models to focus learning on ambiguous regions, significantly improving boundary accuracy [43], [44]. In object detection, uncertainty-based adaptive weighting enhances localization precision by emphasizing hard examples [45]. Similarly, in domain adaptation tasks, uncertainty measurements have been used to assess cross-domain reliability and guide adaptation strategies [46], [47]. These successes demonstrate that uncertainty-guided learning can effectively identify and leverage difficult samples to enhance discriminative capability.

However, despite its effectiveness in other vision tasks, uncertainty-guided learning remains largely underexplored in semisupervised CD. Existing CD methods that incorporate uncertainty primarily use it as an auxiliary signal for pseudolabel filtering or simple loss reweighting [36], [48], rather than as a core guidance principle systematically integrated throughout the learning process. This limited utilization fails to address the unique challenges of CD, where spatial heterogeneity in difficulty is particularly pronounced due to complex boundary transitions and semantic ambiguities. The effective handling of this spatial heterogeneity calls for a unified treatment of uncertainty across core semisupervised components—including augmentation, feature learning, and pseudolabel refinement—yet current approaches do not explore this integration in a systematic manner. This motivates our work to design a comprehensive uncertainty-guided framework specifically tailored for semisupervised CD, where uncertainty serves as the central organizing principle to coordinate augmentation, contrastive learning, and adaptive supervision.

III. Method

A. Overall Framework

The semisupervised CD task is formalized as follows: given a labeled dataset $\mathcal{D}_L = \{(X_i^{t1}, X_i^{t2}, Y_i)\}_{i=1}^{N_L}$ and an unlabeled dataset $\mathcal{D}_U = \{(X_j^{t1}, X_j^{t2})\}_{j=1}^{N_U}$, where $(X_i^{t1}, X_i^{t2}, Y_i)$ represents the ith pair of labeled bitemporal images and their corresponding ground truth (GT) change labels, and (X_j^{t1}, X_j^{t2}) represents the jth pair of unlabeled bitemporal images.

Our proposed USCD method adopts a teacher–student framework [49], where the student model f_s is responsible for primary learning and the teacher model f_t provides stable prediction guidance. Both models employ the same architecture, using ResNet-50 [50] as the encoder and DeepLab [51] as the decoder. The teacher network parameters are updated through exponential moving average (EMA) [52]

$$\theta_t = \alpha\theta_t + (1-\alpha)\,\theta_s \tag{1}$$

where θ_t and θ_s denote the parameters of the teacher and student models, respectively, and α is the momentum coefficient, set to 0.999. For any image pair (X^{t1}, X^{t2}), we define the pixel-level uncertainty measure as follows:

$$U(x,y) = 1 - \left|P_t^0(x,y) - P_t^1(x,y)\right| \tag{2}$$

where $P_t^0(x,y)$ and $P_t^1(x,y)$ represent the prediction probabilities for unchanged and changed classes at position (x,y), respectively.

Fig. 2 illustrates the overall architecture of the USCD framework. The entire framework revolves around the core idea of uncertainty guidance, achieving precise identification and reinforced learning of difficult regions through three collaborative modules. The UAPA module dynamically identifies high-uncertainty regions based on the uncertainty map $U(x,y)$ and implements protective strategies during data augmentation (see Fig. 3). The DRCL module specifically designs contrastive learning mechanisms for difficult regions, enhancing feature discriminability through local and global contrastive losses (see Fig. 4). The UGLR module implements differentiated loss reweighting: high-uncertainty regions in labeled data receive larger weights to strengthen learning, while high-uncertainty regions in unlabeled data receive smaller weights to reduce noise effects (see Fig. 5).

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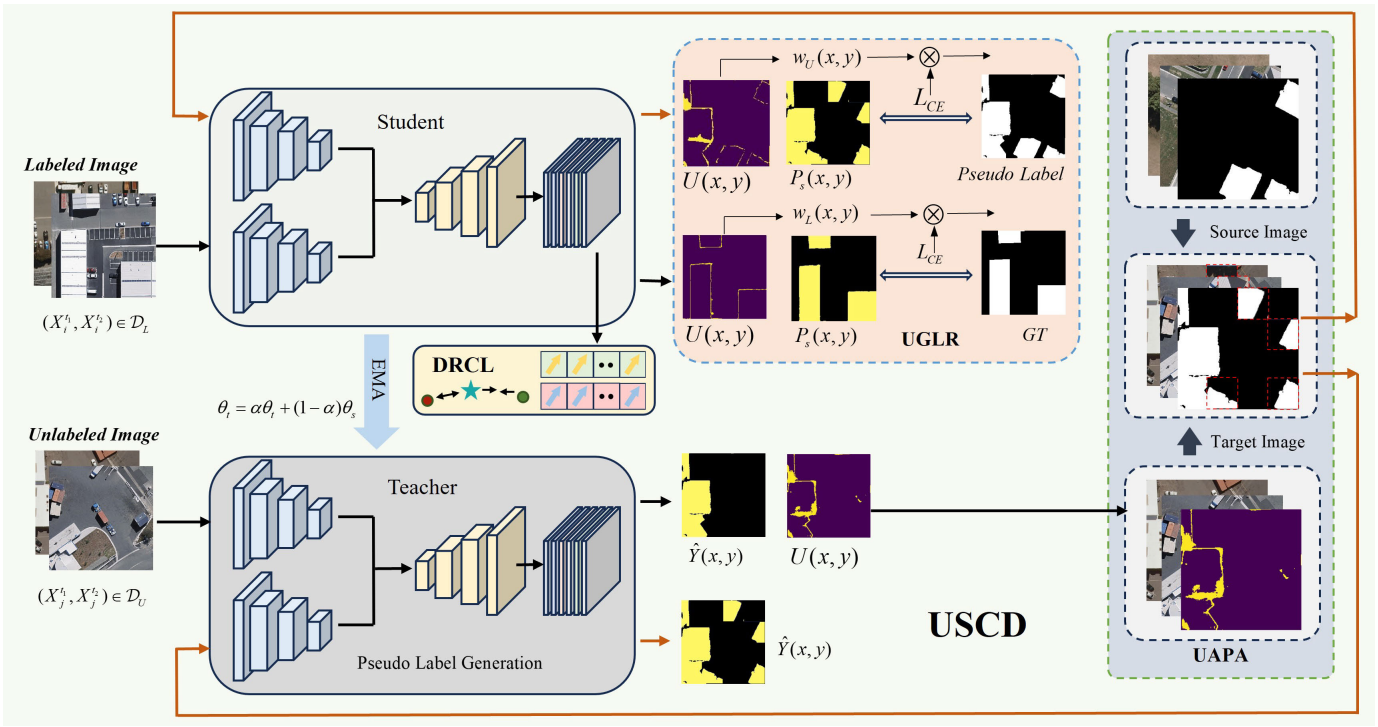


Fig. 2. Overall architecture of the proposed USCD framework. The framework adopts a teacher–student architecture where the student model learns from both labeled and unlabeled data through three synergistic components: 1) UAPA dynamically identifies and protects high-uncertainty regions during data augmentation based on window-level uncertainty scores, with source and target image pairs used to generate augmented mixed pairs guided by the uncertainty map; 2) DRCL enhances feature discriminability in challenging areas through local anchor-based mining and global prototype-based contrastive learning; and 3) UGLR applies differentiated exponential weighting—strengthening supervision in high-uncertainty labeled regions while suppressing noise from unreliable pseudolabels in unlabeled data. The teacher model provides stable predictions and is updated via EMA.

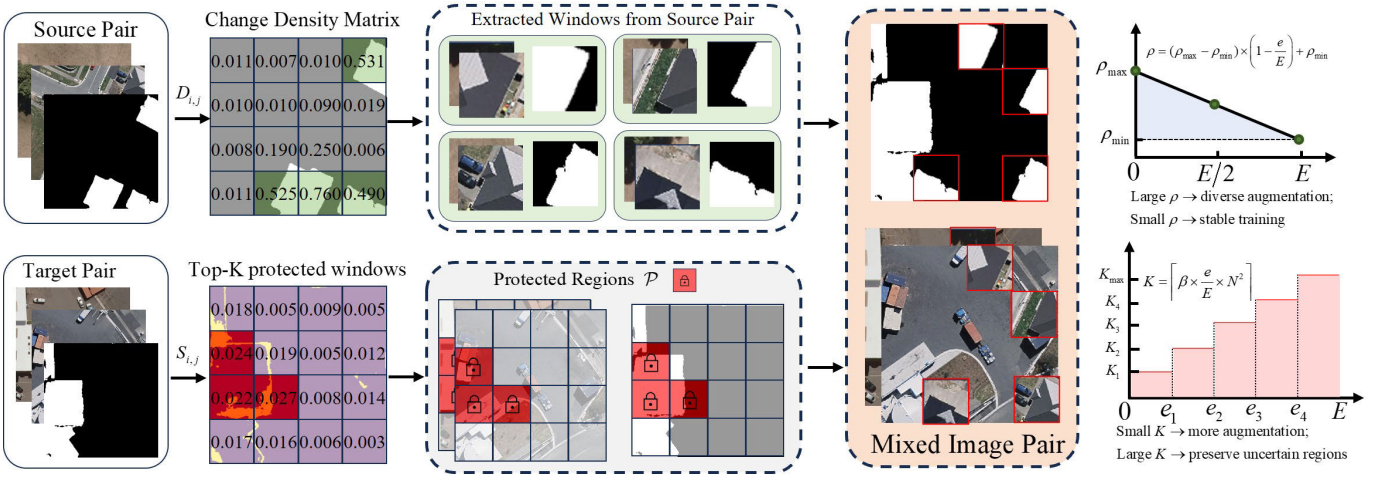


Fig. 3. Illustration of the UAPA module. The source pair provides high-density change windows (top, selected via $D_{i,j}$), while the target pair identifies protected high-uncertainty windows (bottom, selected via $S_{i,j}$). Windows from the source are copy-pasted into nonprotected regions of the target to produce the mixed image pair. (Right) Dynamic schedules of the paste ratio ρ (linearly decreasing) and the number of protected windows K (progressively increasing) across training epochs.

B. Uncertainty-Aware Protective Augmentation

In semisupervised CD, data augmentation is crucial for improving model generalization. We propose a UAPA strategy that preserves key information while maintaining data diversity. Fig. 3 illustrates the overall pipeline of the proposed UAPA module.

1) Window-Based Difficulty Computation: Given teacher network predictions P_t, uniformly divide the feature map into $N \times N$ nonoverlapping windows $W_{i,j}$. For window $W_{i,j}$, the uncertainty score is defined as follows:

$$S_{i,j} = \frac{1}{|W_{i,j}|} \sum_{(x,y) \in W_{i,j}} U(x,y) \tag{3}$$

where $|W_{i,j}|$ denotes the number of pixels in the window.

2) Protected Region Selection: Adopt a dynamic protection strategy that adapts to training progress, where the number of

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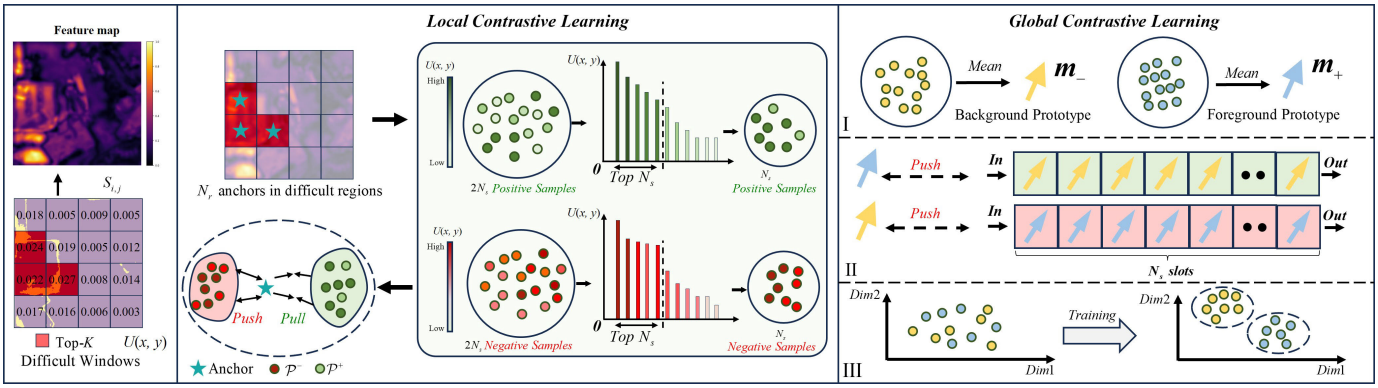


Fig. 4. Architecture of the DRCL module. The module enhances feature discriminability through dual-level contrastive learning. Local level performs anchor-based hard sample mining within difficult regions, selecting top-N_s highest uncertainty samples from reliable foreground/background sets for contrastive learning. Global level maintains prototype memory banks and contrasts current batch prototypes with historical prototypes from opposite classes.

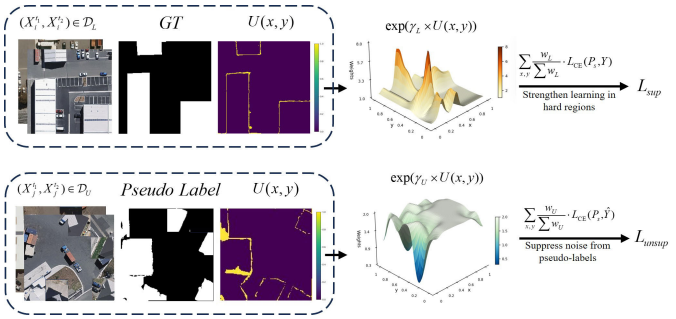


Fig. 5. Illustration of the UGLR module. (Top) For labeled data, the exponential weight $w_L = \exp(\gamma_L \times U)$ with $\gamma_L > 0$ assigns larger weights to high-uncertainty regions, strengthening supervision at difficult boundaries. (Bottom) For unlabeled data, the weight $w_U = \exp(\gamma_U \times U)$ with $\gamma_U < 0$ suppresses contributions from high-uncertainty regions to reduce noise propagation from unreliable pseudolabels.

protected regions K is updated as follows:

$$K = \left\lceil \beta \times \frac{e}{E} \times N^2 \right\rceil \tag{4}$$

where e and E represent the current and total training epochs, respectively, β controls the protection ratio, which is empirically set to 0.3 to balance between preserving difficult regions and maintaining augmentation diversity, and N^2 is the total number of windows. As shown in Fig. 3 (right), K increases progressively with training epochs, ensuring more uncertain regions are preserved as the model matures. We select the top-K windows with the highest uncertainty scores as protected regions

$$\mathcal{I}_K = \arg{*}topK_{i,j} S_{i,j}, \quad \mathcal{P} = \left\{ W_{i,j} \middle| (i,j) \in \mathcal{I}_K \right\}. \tag{5}$$

3) Copy-Paste Source Region Selection: For copy-paste augmentation, we select source regions containing rich change information from nonprotected areas. First, we define the available window set

$$\mathcal{A} = \left\{ W_{i,j} : W_{i,j} \notin \mathcal{P} \right\}. \tag{6}$$

Compute the average change probability for each available window

$$D_{i,j} = \frac{1}{\left| W_{i,j} \right|} \sum_{(x,y) \in W_{i,j}} P_t^1 (x,y). \tag{7}$$

The number of paste windows N_w is dynamically determined as follows:

$$N_w = \lfloor \rho \times |\mathcal{A}| \rfloor \tag{8}$$

where ρ varies with training progress

$$\rho = (\rho_{\max} - \rho_{\min}) \times \left(1 - \frac{e}{E}\right) + \rho_{\min} \tag{9}$$

$\rho_{\max}$ and $\rho_{\min}$ control the range of paste ratios, set to 0.5 and 0.1, respectively. As illustrated in Fig. 3 (top right), ρ decays linearly from $\rho_{\max}$ to $\rho_{\min}$ over training, reflecting a schedule that gradually reduces the number of pasted regions as training progresses, balancing data diversity in early stages with stability in later stages.

Select the top-M windows with the highest change density as copy sources

$$\mathcal{I}_M = \arg{*}topM_{i,j} D_{i,j}, \quad \mathcal{C} = \left\{ W_{i,j} \middle| (i,j) \in \mathcal{I}_M \right\}. \tag{10}$$

4) Adaptive Mixed Augmentation: Given unlabeled image pairs (X_a, X_b) and their pseudolabels (Y_a, Y_b), we perform protective copy-paste operations

$$X_{\text{mix}} = M \odot X_a + (1 - M) \odot X_b \tag{11}$$

$$Y_{\text{mix}} = M \odot Y_a + (1 - M) \odot Y_b \tag{12}$$

where the binary patch mask $M \in \{0,1\}^{N \times N}$ is generated according to:

1) *For Protected Regions ($W_{i,j} \in \mathcal{P}$):* $M_{i,j} = 1$ to preserve high-uncertainty areas containing critical discriminative information.
2) *For Nonprotected Regions ($W_{i,j} \notin \mathcal{P}$):* $M_{i,j}$ is determined by copy-paste operations from high-density change regions to enhance data diversity.

This selective augmentation strategy ensures that difficult regions identified by uncertainty analysis remain intact, preventing the loss of valuable training signals, while still introducing sufficient variation through copy-paste in less critical areas.

C. Difficult Region Contrastive Learning

While generic contrastive frameworks [53], [54] apply supervision uniformly across all spatial locations with random negative sampling, this strategy is insufficient for CD,

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where semantic ambiguity is spatially concentrated in high-uncertainty regions (e.g., building boundaries and visually similar land covers). To address this, DRCL differs from generic designs in three key aspects: it restricts contrastive supervision to high-uncertainty difficult regions; it replaces random sampling with uncertainty-guided hard sample mining; and it filters unreliable anchors via prediction consistency to prevent error propagation. Together, these three task-specific designs distinguish DRCL from generic contrastive learning frameworks and make it better suited to handling semantic ambiguity in CD.

To enhance feature discriminability in difficult regions, we design a contrastive learning mechanism that operates at both local and global levels. A critical prerequisite is ensuring the reliability of training samples—using noisy or unstable predictions as anchors would propagate errors and degrade feature representations.

To address this, we identify reliable regions through prediction consistency. We input both the original image and its UAPA-augmented counterpart into the teacher network, and mark pixels as reliable when their predictions remain consistent across perturbations

$$R(x,y) = \mathbb{I}\left[P_t^{ori}(x,y) = P_t^{aug}(x,y)\right]. \tag{13}$$

This consistency criterion serves as a confidence filter: pixels with stable predictions under augmentation are deemed trustworthy, filtering out unreliable samples that may arise from model uncertainty or augmentation artifacts.

The complete DRCL architecture is illustrated in Fig. 4, which enhances feature learning through dual-level contrastive strategies as described below.

1) Local Contrastive Learning: After extracting feature representations from the student network, we construct foreground and background sample sets $\mathcal{F}_+$ and $\mathcal{F}_-$ from reliable pixels

$$\mathcal{F}_+ = \{f_s : R = 1, y = 1\}, \quad \mathcal{F}_- = \{f_s : R = 1, y = 0\}. \tag{14}$$

We randomly select N_r reliable pixels as anchors within difficult regions. For each anchor, we select $2N_s$ candidate positive samples from the same class and $2N_s$ candidate negative samples from the different class, then use the $\mathcal{T}_{Ns}(\cdot)$ operation (selecting the top N_s samples with highest uncertainty) to construct the final positive and negative sample pairs

$$\mathcal{P}^+ = \mathcal{T}_{Ns}(\mathcal{S}(\mathcal{F}_c, 2N_s), U), \quad \mathcal{P}^- = \mathcal{T}_{Ns}(\mathcal{S}(\mathcal{F}_{\bar{c}}, 2N_s), U) \tag{15}$$

where $\mathcal{S}(\mathcal{F}_c, 2N_s)$ denotes random sampling of $2N_s$ samples from the class-c feature set $\mathcal{F}_c$ and $\bar{c}$ represents the complementary class (i.e., if c is the changed class, then $\bar{c}$ is the unchanged class, and vice versa). This uncertainty-guided hard sample mining ensures that contrastive learning focuses on the most ambiguous samples within difficult regions.

2) Global Contrastive Learning: We maintain two first-in–first-out memory banks to store foreground and background prototypes from difficult regions. For the current batch, we calculate class prototype vectors $\mathbf{m}$ for reliable pixels within difficult regions (superscript h denotes difficult regions)

$$\mathbf{m}_- = \frac{1}{\left|\mathcal{F}_-^h\right|}\sum_{f\in\mathcal{F}_-^h} f, \quad \mathbf{m}_+ = \frac{1}{\left|\mathcal{F}_+^h\right|}\sum_{f\in\mathcal{F}_+^h} f. \tag{16}$$

Each class selects N_s historical prototypes from the memory bank as negative samples for contrastive learning with the current batch prototypes, encouraging class-level separation in feature space.

Both local and global contrastive learning adopt the InfoNCE loss [55] function $\mathcal{L}_{ce}$, where $\mathbf{q}$ represents the query vector

$$\mathcal{L}_{ce} = -\log\frac{\sum_{p^+\in\mathcal{P}^+} e\left(\mathbf{q}, p^+\right)}{\sum_{p^+\in\mathcal{P}^+} e\left(\mathbf{q}, p^+\right) + \sum_{p^-\in\mathcal{P}^-} e\left(\mathbf{q}, p^-\right)}. \tag{17}$$

The final total contrastive loss is: $\mathcal{L}_{contrast} = \mathcal{L}_{local} + 0.5\mathcal{L}_{global}$.

D. Uncertainty-Guided Loss Reweighting

In semisupervised learning, labeled data has reliable GT annotations and should strengthen learning in uncertainty regions; while unlabeled data relies on pseudolabels, which have lower quality in uncertainty regions and should weaken their influence to avoid error propagation. Fig. 5 illustrates the proposed differentiated weighting strategy for labeled and unlabeled data.

Based on the above analysis, we design distinct exponential uncertainty weight functions for labeled and unlabeled data. For pixel position (x, y), the weight functions are defined as follows:

$$w_L(x,y) = \exp(\gamma_L \times U(x,y)) \tag{18}$$

$$w_U(x,y) = \exp(\gamma_U \times U(x,y)) \tag{19}$$

where $\gamma_L > 0$ ensures that high-uncertainty regions in labeled data receive larger weights, while $\gamma_U < 0$ ensures that low-uncertainty regions in unlabeled data receive larger weights to suppress noise propagation from unreliable pseudolabels.

For labeled data, the supervised loss is defined as follows:

$$\mathcal{L}_{sup} = \sum_{x,y}\frac{w_L(x,y)}{\sum_{x',y'} w_L(x',y')}\cdot\mathcal{L}_{CE}(P_s(x,y), Y(x,y)). \tag{20}$$

For unlabeled data, the unsupervised loss is defined as follows:

$$\mathcal{L}_{unsup} = \sum_{x,y}\frac{w_U(x,y)}{\sum_{x',y'} w_U(x',y')}\cdot\mathcal{L}_{CE}\left(P_s(x,y), \hat{Y}(x,y)\right) \tag{21}$$

where $\hat{Y}(x,y)$ denotes the pseudolabel generated by the teacher network.

E. Total Loss Function

The total loss function is defined as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{sup} + \mathcal{L}_{unsup} + 0.1\mathcal{L}_{contrast} \tag{22}$$

where $\mathcal{L}_{sup}$ represents the uncertainty-weighted supervised loss, $\mathcal{L}_{unsup}$ represents the unsupervised loss based on reliable pseudolabels, and $\mathcal{L}_{contrast}$ represents the DRCL loss.

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Algorithm 1 USCD Training Algorithm

Require: Labeled data $\mathcal{D}_L$, Unlabeled data $\mathcal{D}_U$, Student f_s, Teacher f_t
Ensure: Trained change detection model f_s
Initialize f_s and f_t with identical weights
Initialize memory banks $\mathcal{M}^+ \leftarrow \emptyset$, $\mathcal{M}^- \leftarrow \emptyset$
for each training epoch **do**
  for each mini-batch from $\mathcal{D}_L \cup \mathcal{D}_U$ **do**
    // Uncertainty Computation & UAPA
    $P_t \leftarrow f_t(X)$; $U(x,y) \leftarrow 1 - |P_t^0(x,y) - P_t^1(x,y)|$
    $\mathcal{P} \leftarrow$ TopK(WindowUncertainty(U)) {Protected regions}
    $(X_{aug}, \hat{Y}) \leftarrow$ CopyPaste($X, \mathcal{P}$) {With protection}
    // DRCL: Local & Global Contrastive Learning
    $\mathcal{F}_s \leftarrow f_s$.encoder($X_{aug}$) {Extract features}
    Build reliable sets: $\mathcal{F}^+ \leftarrow \{f : R = 1, y = 1\}$, $\mathcal{F}^- \leftarrow \{f : R = 1, y = 0\}$
    $\mathcal{L}_{local} \leftarrow \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_{InfoNCE}(q_k, \text{HardPos}_k, \text{HardNeg}_k)$
    $\mathbf{m}^+ \leftarrow \text{mean}(\mathcal{F}^+)$; $\mathbf{m}^- \leftarrow \text{mean}(\mathcal{F}^-)$ {Prototypes}
    Update: $\mathcal{M}^+ \leftarrow \mathcal{M}^+ \cup \{\mathbf{m}^+\}$; $\mathcal{M}^- \leftarrow \mathcal{M}^- \cup \{\mathbf{m}^-\}$
    $\mathcal{L}_{global} \leftarrow \mathcal{L}_{InfoNCE}(\mathbf{m}^+, \mathcal{M}^-) + \mathcal{L}_{InfoNCE}(\mathbf{m}^-, \mathcal{M}^+)$
    $\mathcal{L}_{contrast} \leftarrow \mathcal{L}_{local} + 0.5\mathcal{L}_{global}$
    // UGLR: Uncertainty-guided Loss Re-weighting
    $P_s \leftarrow f_s(X_{aug})$ {Student prediction}
    $w_L \leftarrow \exp(2.0 \times U)$; $w_U \leftarrow \exp(-1.0 \times U)$ {Adaptive weights}
    $\mathcal{L}_{sup} \leftarrow \sum w_L \cdot \mathcal{L}_{CE}(P_s^L, Y)$
    $\mathcal{L}_{unsup} \leftarrow \sum w_U \cdot \mathcal{L}_{CE}(P_s^U, \hat{Y})$
    // Update
    $\mathcal{L}_{total} \leftarrow \mathcal{L}_{sup} + \mathcal{L}_{unsup} + 0.1\mathcal{L}_{contrast}$
    $\theta_s \leftarrow \theta_s - \eta\nabla_{\theta_s}\mathcal{L}_{total}$ {Update student}
    $\theta_t \leftarrow 0.999\theta_t + 0.001\theta_s$ {Update teacher (EMA)}
  end for
end for
return: f_s

F. Complete Training Algorithm

Algorithm 1 presents the complete training procedure of our USCD framework.

IV. Experiment

A. Datasets and Implementation Details

1) Datasets: To comprehensively evaluate the effectiveness and generalization capability of our proposed USCD framework, we conduct extensive experiments on six diverse benchmark datasets spanning different scenarios, image platforms, and change characteristics. Table I summarizes the key properties of these datasets, including spatial resolution, image size, data splits, temporal spans, and change types.

1) *LEVIR-CD [56]:* A large-scale remote sensing building CD dataset containing 637 pairs of high-resolution images (0.5 m/pixel, 1024×1024 pixels). After preprocessing, the images are divided into 256×256 pixel patches, resulting in 7120 pairs for training, 1024 pairs for validation, and 2048 pairs for testing.
2) *WHU-CD:* A building CD dataset constructed from aerial images. Covering an area of 20.5 km^2, it records building changes due to postearthquake reconstruction (from 12 796 to 16 077 buildings). The dataset is divided into 5947 pairs for training, 743 pairs for validation, and 744 pairs for testing (256×256 pixels).
3) *CDD [57]:* A seasonal CD dataset containing 16 000 images collected from Google Earth with pixel resolution ranging from 0.03 to 1 m. The dataset is divided into 256×256 pixel patches, including 10 000 pairs for training, 3000 pairs for validation, and 3000 pairs for testing.
4) *S2Looking [58]:* A large-scale building CD dataset containing 5000 registered bitemporal image pairs of size 1024×1024 pixels. The images are sourced from various satellite platforms with spatial resolutions ranging from 0.5 to 0.8 m. Following the standard protocol, images are cropped into 256×256 pixel patches, resulting in multiple training, validation, and test pairs covering diverse geographical regions.
5) *SYSU-CD [59]:* An urban CD dataset comprising 20 000 pairs of 256×256 aerial image patches at 0.5-m spatial resolution, covering approximately 744 km^2 in Hong Kong. The dataset captures various urban land-cover changes, including building construction and vegetation dynamics. The 20 000 image pairs are split into 12 000 training pairs, 4000 validation pairs, and 4000 test pairs.
6) *JL1-CD [60]:* A large-scale submeter resolution CD benchmark derived from Jilin-1 satellite imagery. The dataset exhibits diverse change types with varying change area ratios (CARs), capturing building changes, urban development, and land-cover transformations in rapidly evolving regions. Images are processed into standard patches for training, validation, and testing following conventional protocols.

2) Implementation Details: Our experiments are implemented based on the PyTorch framework and trained on an NVIDIA Tesla A100 GPU. The training strategy adopts a total of 100 epochs, where the first 30 epochs use fully supervised training with only labeled data to establish basic CD capabilities. The optimization configuration includes a batch size of 8, an initial learning rate of 0.01 with a linear decay strategy, gradually reducing to $1e^{-4}$. The momentum parameter is set to 0.9, and the weight decay coefficient is $1e^{-4}$. To enhance the model's generalization ability, we employ a basic weak augmentation pipeline for labeled data, including random horizontal flipping, random rotation, and random cropping. To ensure fair comparison, all competing methods are trained with the same basic labeled data augmentation pipeline. We set the control parameters for UGLR as $\gamma_L = 2.0$ and $\gamma_U = -1.0$, the temperature parameter for contrastive learning as $\tau = 0.1$, the number of anchors in difficult regions as $N_r = 32$, and the number of positive and negative samples for each anchor as $N_s = 64$. The confidence threshold for pseudolabel generation is set to 0.9. We set window number N = 16 for S2Looking, N = 4 for SYSU-CD, and N = 8 for the remaining four datasets.

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TABLE I
SUMMARY OF CD DATASETS USED IN OUR EXPERIMENTS. CR DENOTES THE CHANGE RATIO, DEFINED AS THE PROPORTION OF CHANGED PIXELS IN THE TRAINING LABELS

Datasets	Spatial Resolution	Patch Size	Train/Val/Test	Time Spans	Change Types	CR (%)	Download
LEVIR-CD	0.5m	256×256	7120/1024/2048	5-14 years	Building	4.59	Link
WHU-CD	0.2m	256×256	5947/743/744	2012-2016	Building	4.38	Link
CDD	0.03-1.0m	256×256	10000/3000/3000	Varying	Seasonal	12.81	Link
S2Looking	0.5-0.8m	256×256	56000/8000/16000	Varying	Building	1.21	Link
SYSU-CD	0.5m	256×256	12000/4000/4000	2007-2014	Urban	21.34	Link
JL1-CD	0.5-0.75m	256×256	16000/2000/2000	Varying	Building/Urban	9.83	Link

Oracle Baseline: To evaluate the effectiveness of our semisupervised approach, we establish an oracle baseline that represents the performance upper bound. The oracle model is trained using standard fully supervised learning with 100% of the labeled training data, employing the same network architecture (ResNet-50 + DeepLab) and training configuration as our semisupervised framework, but without any components (i.e., no teacher–student framework, no pseudolabeling, and no uncertainty-guided modules). The oracle serves as a reference to measure how close our method can approach the fully supervised performance while using only a fraction of labeled data.

3) Evaluation Metrics: IoU is used to measure the accuracy of change prediction

$$IoU = \frac{TP}{TP + FP + FN}. \tag{23}$$

$F1$-Score serves as the harmonic mean of precision and recall, providing a more comprehensive performance evaluation

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{24}$$

where TP, FP, and FN represent the number of true positive, false positive, and false negative pixels, respectively.

B. Comparison With State-of-the-Art Methods

1) Quantitative Results: Table II presents quantitative comparisons of our method with state-of-the-art semisupervised CD methods across six benchmark datasets, including AdvEnt [61], s4GAN [62], SemiCDNet [34], SemiCD [33], SemiCD-VL [63], UniMatch [64], CutMix-CD [65], and PUF [36]. These datasets cover diverse CD scenarios, including building CD (WHU-CD, LEVIR-CD), seasonal change detection, multiscale satellite imagery (S2Looking), urban area monitoring (SYSU), and high-resolution satellite CD (JL1-CD).

Our USCD method demonstrates consistent superiority across all datasets and annotation ratios. With only 5% labeled data, we achieve significant improvements over the second-best methods: +0.61% $F1$ on WHU-CD (89.41% versus 88.80%), +0.21% $F1$ on LEVIR-CD (90.22% versus 90.01%), +0.96% $F1$ on CDD (87.27% versus 86.31%), +1.10% $F1$ on S2Looking (55.99% versus 54.89%), +0.72% $F1$ on SYSU (79.94% versus 79.22%), and +1.35% $F1$ on JL1-CD (71.15% versus 69.80%). These improvements are particularly meaningful considering the extremely limited supervision, validating that our uncertainty-guided learning strategy effectively identifies and focuses on difficult regions to maximize learning efficiency.

The performance advantages become more pronounced on challenging datasets. On S2Looking and JL1-CD, where complex multiscale objects and diverse change patterns exist, our method shows substantial gains over Sup. only (baseline using only the specified percentage of labeled data) 13.84% and 16.88% $F1$, respectively, with 5% annotations, demonstrating the critical value of semisupervised learning when annotations are scarce, and data characteristics make CD inherently difficult. On the seasonal CD dataset CDD, our method achieves 87.27% $F1$ with 5% labels, effectively distinguishing real changes from seasonal variations through enhanced feature discriminability in uncertain regions.

As annotation ratios increase to 10%, 20%, and 40%, our method generally maintains leading or highly competitive performance across all datasets. Notably, on WHU-CD and LEVIR-CD with 40% annotations, we approach oracle performance with only 1.72% and 0.68% $F1$ gaps, respectively, demonstrating that our uncertainty-guided approach efficiently reduces annotation requirements while maintaining near-oracle performance. The consistent improvements across diverse scenarios—including variations in spatial resolution (0.03–1 m), image platforms (aerial versus satellite), and change types—validate the robustness and generalization capability of our uncertainty-guided learning paradigm.

2) Qualitative Results: Fig. 6 presents qualitative comparison results across multiple datasets with 20% labeled data. The supervised baseline exhibits considerable false alarms (shown in red) and missed detections (shown in green), particularly struggling with building boundaries and small change regions. Existing semisupervised methods show improvements but still suffer from detection errors: s4GAN produces scattered false positives in complex backgrounds, SemiCDNet misses subtle boundary changes, and even stronger methods like UniMatch and CutMix-CD exhibit limitations in precisely delineating change boundaries.

Our USCD method significantly reduces both false positives and false negatives across all scenarios. The detection results show cleaner boundaries, more complete change region coverage, and fewer scattered errors. Particularly in challenging cases involving complex backgrounds, small changes, or subtle boundaries, USCD demonstrates superior accuracy. For example, on LEVIR-CD, our method accurately captures complete

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TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART SEMISUPERVISED CD METHODS ON SIX DATASETS

Method	WHU-CD								Method	LEVIR-CD							
	5%		10%		20%		40%			5%		10%		20%		40%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU		F1	IoU	F1	IoU	F1	IoU	F1	IoU
Sup. only	72.54	60.72	77.25	65.96	80.69	70.32	82.44	77.69	Sup. only	69.56	49.95	80.24	68.26	83.45	72.97	86.59	76.23
AdvEnt	78.63	64.79	82.47	70.17	85.26	74.31	86.68	76.48	AdvEnt	82.98	70.91	84.74	73.52	84.90	73.77	85.00	73.91
s4GAN	74.30	59.11	77.07	62.70	82.50	70.22	85.59	74.81	s4GAN	78.41	64.49	79.53	66.02	84.14	72.62	85.03	73.96
SemiCDNet	81.48	68.75	83.79	72.11	85.32	74.40	86.84	76.79	SemiCDNet	82.69	70.48	85.00	73.91	85.12	74.10	85.38	75.40
SemiCD	79.36	65.79	80.97	68.03	85.42	74.55	87.65	78.01	SemiCD	85.18	74.19	87.08	77.12	87.55	77.85	88.26	78.99
SemiCD-VL	87.55	80.56	89.49	82.56	90.69	83.01	91.36	83.99	SemiCD-VL	87.64	80.74	89.37	81.88	90.79	82.26	91.27	82.99
UniMatch	88.61	79.54	89.96	81.31	90.01	81.84	91.77	84.79	UniMatch	89.43	80.88	89.95	81.73	90.13	82.04	90.26	82.25
CutMix-CD	88.80	81.48	90.46	82.59	91.14	83.27	92.00	85.18	CutMix-CD	90.01	81.48	90.32	82.34	90.43	82.53	91.06	83.58
PUF	88.79	81.01	90.01	81.99	90.82	82.84	91.41	84.27	PUF	89.21	80.70	90.08	82.21	90.41	82.61	90.94	83.01
Ours	**89.41**	**81.52**	**90.97**	**83.01**	**91.22**	**83.79**	**92.97**	**85.89**	**Ours**	**90.22**	**81.56**	**90.99**	**82.55**	**91.21**	**82.99**	**91.91**	**83.79**
Oracle: F1=94.69, IoU=88.36									Oracle: F1=92.59, IoU=84.52								

Method	CDD								Method	S2Looking							
	5%		10%		20%		40%			5%		10%		20%		40%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU		F1	IoU	F1	IoU	F1	IoU	F1	IoU
Sup. only	75.32	59.21	79.43	66.43	85.42	77.48	89.91	81.03	Sup. only	42.15	26.78	47.79	31.40	57.93	40.78	61.76	44.67
AdvEnt	72.94	57.40	79.98	66.64	85.12	75.79	89.13	80.39	AdvEnt	43.89	28.12	49.23	32.67	58.76	41.54	62.52	45.41
s4GAN	72.94	57.40	79.98	66.64	87.37	77.58	90.06	81.91	s4GAN	43.23	27.65	48.65	32.18	58.34	41.23	62.08	44.95
SemiCDNet	76.74	62.27	83.42	71.56	87.94	78.47	89.57	81.10	SemiCDNet	41.34	26.12	46.79	30.54	55.28	38.19	59.84	42.69
SemiCD	78.24	64.26	84.99	73.90	89.09	80.32	90.77	83.10	SemiCD	46.78	30.56	51.67	34.78	59.89	42.67	63.15	46.23
SemiCD-VL	85.33	77.26	89.76	81.01	91.32	85.05	93.21	88.01	SemiCD-VL	49.92	33.45	54.89	37.92	61.78	44.89	65.12	48.34
UniMatch	85.56	77.47	89.68	81.29	92.64	86.29	94.46	89.51	UniMatch	45.67	29.78	50.52	33.80	60.69	43.57	63.20	46.20
CutMix-CD	86.26	77.96	**91.05**	**83.57**	**93.84**	**88.39**	94.92	90.34	CutMix-CD	54.89	38.23	60.57	43.44	63.44	46.45	65.92	49.17
PUF	86.31	77.94	89.40	80.80	92.40	86.30	93.90	90.20	PUF	53.67	37.45	59.23	42.15	62.78	45.89	65.34	48.76
Ours	**87.27**	**79.02**	90.77	82.35	93.49	86.97	**95.69**	**90.91**	**Ours**	**55.99**	**39.39**	**61.70**	**44.72**	**64.83**	**47.96**	**67.42**	**50.61**
Oracle: F1=96.64, IoU=91.97									Oracle: F1=67.66, IoU=51.33								

Method	SYSU-CD								Method	JL1-CD							
	5%		10%		20%		40%			5%		10%		20%		40%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU		F1	IoU	F1	IoU	F1	IoU	F1	IoU
Sup. only	75.92	61.34	77.46	63.28	77.91	63.89	79.33	65.72	Sup. only	54.27	37.24	69.45	53.20	73.22	57.76	76.45	62.34
AdvEnt	76.81	62.02	78.22	64.10	78.79	64.72	80.11	66.86	AdvEnt	59.03	41.87	67.24	50.65	71.14	55.21	74.67	59.45
s4GAN	73.19	57.88	74.07	58.94	75.33	60.48	76.67	62.26	s4GAN	53.84	36.83	68.78	52.42	74.10	58.86	77.89	63.78
SemiCDNet	71.91	56.20	75.03	60.02	75.42	60.56	76.89	62.56	SemiCDNet	54.51	37.47	68.13	51.66	68.58	52.18	72.34	56.89
SemiCD	78.41	64.63	78.24	64.42	79.70	66.41	81.17	68.60	SemiCD	57.09	39.95	72.88	57.34	75.79	61.02	79.23	65.67
SemiCD-VL	78.11	64.30	79.02	65.34	80.29	66.67	81.78	69.30	SemiCD-VL	67.89	51.39	71.52	55.67	73.49	58.09	76.89	62.45
UniMatch	79.21	64.67	79.38	65.01	80.51	67.03	82.09	69.71	UniMatch	69.80	53.61	72.98	57.46	75.22	60.28	78.67	64.89
CutMix-CD	78.67	64.52	79.11	64.82	80.03	66.54	81.56	69.04	CutMix-CD	65.89	49.13	73.42	58.01	75.40	60.52	78.34	64.56
PUF	79.22	65.80	79.53	66.22	79.67	66.43	81.32	68.78	PUF	62.80	45.78	72.14	56.42	75.77	60.46	78.92	65.12
Ours	**79.94**	**66.83**	**80.62**	**67.45**	**81.53**	**68.35**	**82.99**	**70.81**	**Ours**	**71.15**	**55.23**	**74.67**	**59.78**	**76.89**	**62.15**	**80.34**	**66.89**
Oracle: F1=82.4, IoU=69.6									Oracle: F1=82.54, IoU=70.35								

building footprints with precise boundaries, while comparison methods miss building corners or produce irregular edges. On CDD, USCD successfully distinguishes seasonal vegetation changes from real land cover changes, avoiding false alarms that confuse other methods.

The visual comparisons further demonstrate the effectiveness of our method in producing more accurate and complete change maps. Compared with the baseline and existing semisupervised methods, USCD consistently reduces scattered false alarms and missed detections, especially around building

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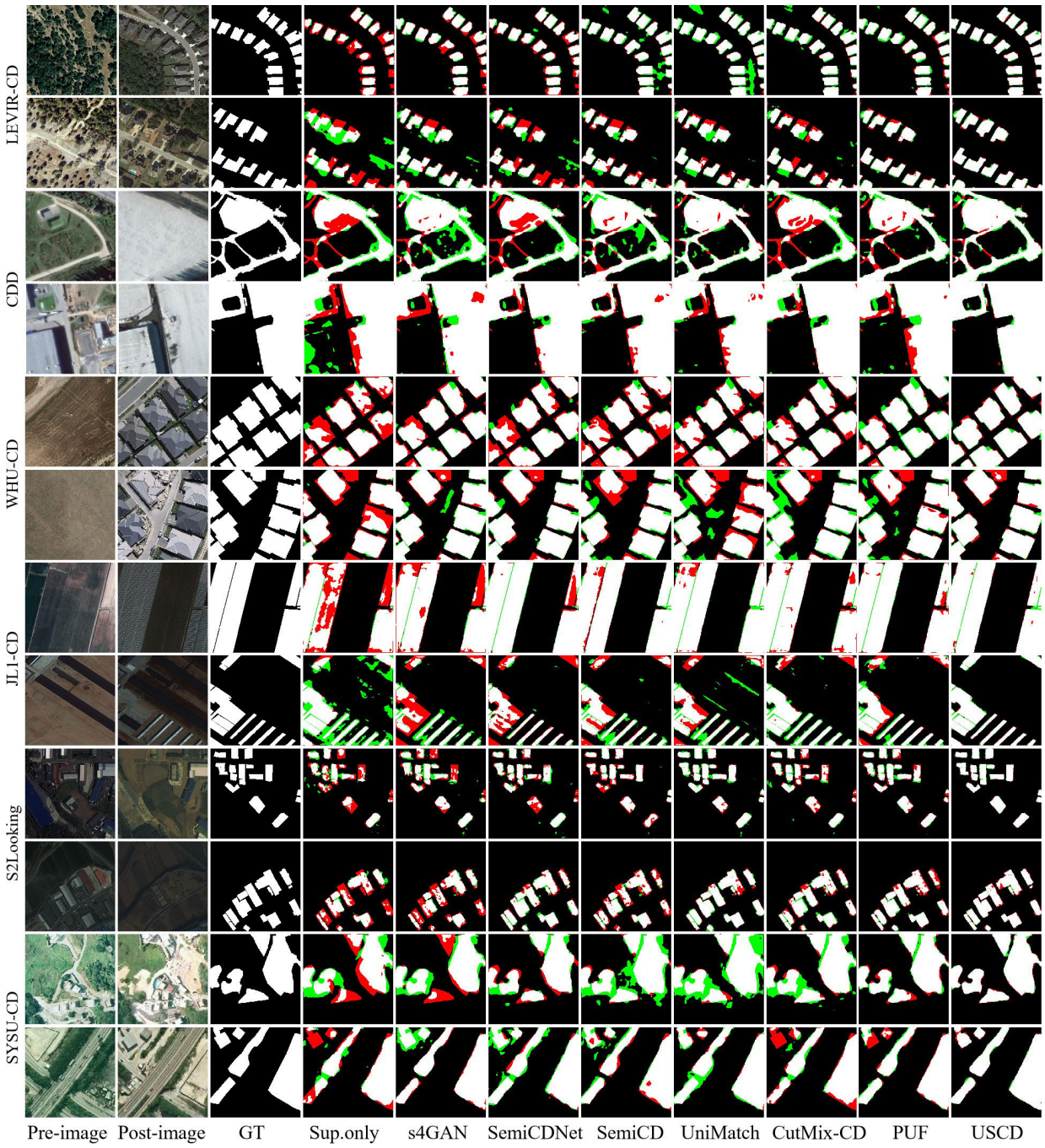


Fig. 6. Qualitative comparison across six datasets with 20% labeled data. Rows represent: LEVIR-CD (1 and 2), CDD (3 and 4), WHU-CD (5 and 6), JL1-CD (7 and 8), S2Looking (9 and 10), and SYSU-CD (11 and 12). Columns: T1/T2 images, GT, and predictions from Sup. only, s4GAN, SemiCDNet, SemiCD, UniMatch, CutMix-CD, PUF, and our USCD. Red represents false positive errors (false alarms) and green represents false negative errors (missed detections).

boundaries, small changed objects, and complex background regions. The predicted change regions are more coherent and better aligned with the GT, indicating that the proposed uncertainty-guided learning strategy enhances the model's ability to distinguish genuinely changed areas from confusing pseudochange patterns across diverse CD scenarios.

C. Ablation Study

Table III presents comprehensive ablation experiments on LEVIR-CD and WHU-CD datasets to validate the effectiveness of each proposed component across different annotation ratios.

1) Main Component Analysis: Starting from the teacher–student baseline, each component contributes progressively to the final performance. On LEVIR-CD with 5% labels, UAPA provides the largest improvement (+1.81% $F1$, from 86.34% to 88.15%), demonstrating that protecting uncertain regions during augmentation is crucial for preserving critical information. DRCL further improves performance by +1.30% $F1$ (to 89.45%), validating the

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TABLE III

ABLATION STUDY ON LEVIR-CD AND WHU-CD DATASETS. THE TABLE SHOWS THE CONTRIBUTION OF EACH COMPONENT AND THEIR INTERNAL DESIGN CHOICES ACROSS DIFFERENT ANNOTATION RATIOS

Method	LEVIR-CD								WHU-CD							
	5%		10%		20%		40%		5%		10%		20%		40%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU
Main Ablation Study																
Baseline	86.34	76.92	87.21	78.38	88.56	79.45	89.12	80.23	84.23	72.67	86.45	76.12	87.89	78.45	89.34	80.67
+ UAPA	88.15	78.74	89.32	80.15	89.87	81.01	90.56	82.12	86.78	76.89	88.67	79.56	89.45	80.89	90.89	83.12
+ UAPA + DRCL	89.45	80.12	90.58	81.89	90.76	82.34	91.23	83.12	88.12	78.92	89.78	81.34	90.34	82.45	91.67	84.56
+ Full	**90.22**	**81.56**	**90.99**	**82.55**	**91.21**	**82.99**	**91.91**	**83.79**	**89.41**	**81.52**	**90.97**	**83.01**	**91.22**	**83.79**	**92.97**	**85.89**
UAPA Ablation																
w/o Dynamic	87.56	77.89	88.78	79.67	89.45	80.56	90.12	81.45	85.89	75.23	87.89	78.12	88.78	79.89	90.23	81.89
w/o Protection	86.92	77.23	88.12	79.02	89.01	79.89	89.78	80.89	85.34	74.56	87.23	77.45	88.34	79.23	89.78	81.34
Full UAPA	**88.15**	**78.74**	**89.32**	**80.15**	**89.87**	**81.01**	**90.56**	**82.12**	**86.78**	**76.89**	**88.67**	**79.56**	**89.45**	**80.89**	**90.89**	**83.12**
DRCL Ablation																
Local Only	88.89	79.45	90.12	81.23	90.34	81.78	90.89	82.56	87.67	77.89	89.34	80.67	89.89	81.67	91.23	84.01
Global Only	88.67	79.12	89.87	80.98	90.12	81.45	90.67	82.23	87.45	77.56	89.12	80.34	89.67	81.34	91.01	83.78
Full DRCL	**89.45**	**80.12**	**90.58**	**81.89**	**90.76**	**82.34**	**91.23**	**83.12**	**88.12**	**78.92**	**89.78**	**81.34**	**90.34**	**82.45**	**91.67**	**84.56**
UGLR Ablation																
Uniform Weight	89.12	79.78	90.23	81.34	90.45	81.89	90.89	82.67	87.89	78.34	89.45	80.89	90.12	82.12	91.34	84.23
Labeled Only	89.67	80.45	90.67	82.12	90.89	82.56	91.45	83.34	88.67	79.89	90.23	81.89	90.78	82.89	92.12	85.12
Full UGLR	**90.22**	**81.56**	**90.99**	**82.55**	**91.21**	**82.99**	**91.91**	**83.79**	**89.41**	**81.52**	**90.97**	**83.01**	**91.22**	**83.79**	**92.97**	**85.89**

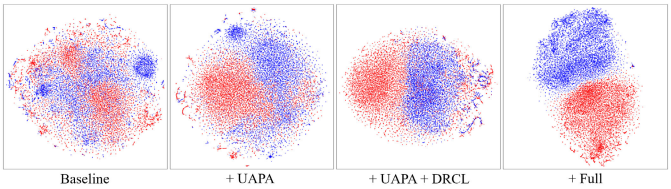


Fig. 7. t-SNE visualization of learned feature representations on LEVIR-CD (5% labels). Red and blue points represent unchanged and changed classes, respectively. Progressive addition of components (UAPA, DRCL, full) leads to increasingly compact intraclass clustering and clearer interclass separation, validating that each module contributes to learning more discriminative features.

importance of enhanced feature learning in difficult regions through contrastive learning. UGLR contributes an additional +0.77% $F1$ (to 90.22%), showing that adaptive loss weighting effectively balances learning between labeled and unlabeled data. Similar trends are observed on WHU-CD, where the three components contribute +2.55%, +1.34%, and +1.29% $F1$ improvements, with 5% annotations. The consistent performance gains across both datasets and all annotation ratios validate the generalization and robustness of our proposed components.

To gain deeper insights into how each component improves feature learning, we visualize the learned representations extracted from the encoder's final layer using t-SNE [66] in Fig. 7. We perform this visualization on the LEVIR-CD dataset with 5% labeled data. The baseline shows significant overlap between changed (blue) and unchanged (red) samples. Adding UAPA reduces the overlap by preserving discriminative information in difficult regions during augmentation. Furthermore, adding DRCL notably tightens intraclass clusters and widens interclass margins through contrastive learning. The full model achieves the most compact and well-separated clusters, demonstrating that all components synergistically contribute to learning more discriminative feature representations.

2) UAPA Component Analysis: The internal ablation of UAPA reveals that both dynamic protection and the protection mechanism itself are essential. Removing dynamic protection results in performance drops of 0.59% and 0.89% $F1$ on LEVIR-CD and WHU-CD, respectively, at 5% annotation while removing the entire protection mechanism causes more severe degradation (1.23% and 1.44% $F1$ drops). These results confirm that uncertainty-guided protective augmentation prevents the loss of critical information in difficult regions, which is particularly important when labeled data is scarce.

3) DRCL Component Analysis: For DRCL, we evaluate the contributions of local and global contrastive learning separately. Using local contrastive learning alone achieves 88.89% and 87.67% $F1$ on LEVIR-CD and WHU-CD, respectively, while global contrastive learning alone yields 88.67% and 87.45% $F1$. Combining both strategies (Full DRCL) improves performance to 89.45% and 88.12% $F1$, demonstrating their complementary effects: local contrastive learning enhances fine-grained feature discriminability within difficult regions,

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TABLE IV
CROSS-DOMAIN GENERALIZATION PERFORMANCE. RESULTS SHOW MODEL PERFORMANCE WHEN TRAINED ON ONE DATASET AND TESTED ON ANOTHER

Method	LEVIR-CD → WHU-CD				LEVIR(Sup.) + WHU(Unsup.) → LEVIR			
	5%		10%		5%		10%	
	F1	**IoU**	F1	**IoU**	F1	**IoU**	F1	**IoU**
Sup-only	45.27	28.96	51.39	36.76	79.09	65.41	81.85	69.27
SemiCDNet	47.81	31.41	55.27	38.19	78.51	64.62	81.62	68.95
RCR	49.31	32.72	65.98	49.23	81.22	68.37	83.04	71.00
RCL [6]	52.15	35.27	58.03	40.87	83.65	71.89	84.03	72.46
FPA [67]	50.73	33.99	66.07	49.34	73.65	58.29	79.95	66.60
UniMatch	52.24	35.65	63.87	46.99	81.76	69.47	83.73	72.45
CutMix-CD	52.97	36.01	64.48	48.35	82.41	69.23	84.02	73.10
Ours	**54.69**	**38.42**	**66.95**	**51.32**	**83.72**	**70.47**	**85.42**	**74.11**

while global contrastive learning provides class-level semantic guidance through prototype consistency.

4) UGLR Component Analysis: The UGLR ablation compares three weighting strategies: uniform weighting, reweighting labeled data only, and the full strategy that differentially weights both labeled and unlabeled data. On LEVIR-CD with 5% labels, uniform weighting achieves 89.12% $F1$, labeled-only reweighting improves to 89.67% $F1$, and the full strategy reaches 90.22% $F1$. Similar trends are observed on WHU-CD with consistent improvements. The results validate our design principle: increasing weights for high-uncertainty regions in labeled data strengthens learning of difficult patterns while decreasing weights for high-uncertainty regions in unlabeled data suppresses noise propagation from unreliable pseudolabels.

D. Cross-Domain Generalization

To evaluate the generalization capability of our method across different domains, we conduct two types of cross-domain experiments as shown in Table IV.

1) Direct Cross-Domain Transfer (LEVIR-CD → WHU-CD): We train on LEVIR-CD and directly test on WHU-CD without any fine-tuning. Our method achieves 54.69% $F1$ and 38.42% IoU with 5% labeled data, outperforming CutMix-CD by 1.72% and 2.41%, respectively. At a 10% annotation ratio, the performance gap widens to 66.95% $F1$ and 51.32% IoU, further validating the superior cross-domain transferability. These results indicate that our uncertainty-guided learning captures more generalizable change patterns across different imaging conditions.

2) Unsupervised Domain Adaptation: We use labeled LEVIR-CD data with unlabeled WHU-CD data for training, then test on LEVIR-CD. Our method achieves 83.72% $F1$ and 70.47% IoU with 5% supervision, demonstrating effective utilization of cross-domain unlabeled data to enhance source domain performance.

The cross-domain results demonstrate two key advantages of our uncertainty-guided approach: 1) *enhanced transferability*—uncertainty quantification helps identify domain-invariant difficult regions crucial for cross-domain generalization, enabling robust performance even with significant domain shifts; and 2) *effective cross-domain data utilization*—our method better leverages unlabeled data from different domains to improve source domain performance.

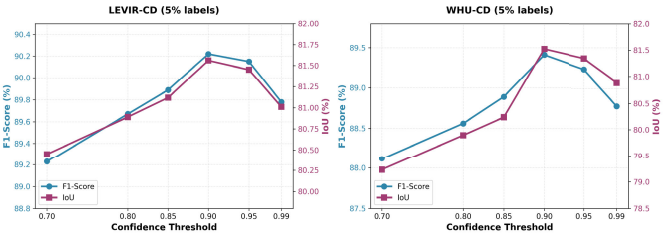


Fig. 8. Sensitivity analysis of confidence threshold on LEVIR-CD and WHU-CD datasets with 5% labeled data. Both $F1$-score (blue) and IoU (purple) metrics are shown.

E. Parameter Sensitivity Analysis

Fig. 8 presents the performance variations across different confidence thresholds ranging from 0.7 to 0.99 on both LEVIR-CD and WHU-CD datasets. The results show that a confidence threshold of 0.90 achieves the best performance on both datasets, reaching 90.22% $F1$-Score and 81.56% IoU on LEVIR-CD, and 89.41% $F1$-Score and 81.52% IoU on WHU-CD. This threshold effectively balances between utilizing sufficient pseudolabels for learning and filtering out unreliable predictions. The performance remains stable within the range [0.85, 0.95], with $F1$-Score variations less than 0.4% on both datasets. This stability demonstrates that our method is not overly sensitive to the exact threshold value, which is important for practical deployment where optimal hyperparameters may vary across different scenarios. When the threshold is too low (0.70), noisy pseudolabels are included, leading to performance drops of approximately 1.0% $F1$-Score. Conversely, excessively high thresholds (0.99) result in insufficient pseudolabeled samples for effective learning, causing similar performance degradation. This validates the importance of proper threshold selection while confirming the robustness of our default setting.

F. Extreme Label Scarcity Analysis

Fig. 9 presents performance comparisons under extreme annotation scarcity. Three key findings emerge.

1) *Supervised Learning Fails Dramatically:* With only 1% labels, the supervised baseline achieves merely 35%–38% IoU, highlighting the critical need for semisupervised approaches.
2) *USCD Excels at Minimal Supervision:* Our method demonstrates maximum advantage when labels are most scarce: +3.3%/+4.7% at 1%, gradually diminishing to +1.7%/+2.2% at 3%. This validates that uncertainty-guided learning is most valuable under severe supervision constraints.
3) *Significant Annotation Savings:* USCD with 1% labels achieves comparable performance to strong baselines with 2% labels, effectively halving annotation requirements while maintaining practical accuracy.

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TABLE V
EFFICIENCY COMPARISON ON LEVIR-CD WITH 5% LABELED DATA

Method	Params.(M)	FLOPs(G)	Epoch (s)	Inference (ms)
s4GAN	40.47	29.10	409.93	38.72
SemiCD	46.85	73.23	692.25	34.74
C2FNet	16.17	62.10	395.33	48.10
RCL	46.85	73.23	477.67	35.34
SemiSANet	1.61	4.67	51.45	28.42
FPA	46.85	73.23	588.67	35.25
CutMix-CD	46.85	73.23	745.95	35.94
SemiCD-VL	24.88	21.60	971.88	44.46
UniMatch	40.47	29.10	386.42	41.97
Ours	**40.47**	**29.10**	**394.53**	**38.20**

TABLE VI
GENERALIZATION EVALUATION OF OUR UNCERTAINTY-GUIDED STRATEGY ON DIFFERENT BASELINE METHODS

Methods	LEVIR-CD				WHU-CD			
	5%		10%		5%		10%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU
FPA	84.94	73.83	85.71	74.75	81.42	68.75	83.24	71.19
FPA + Ours	**86.40**	**75.99**	**87.64**	**76.67**	**83.18**	**71.02**	**84.95**	**73.41**
SemiCD-VL	87.55	80.56	88.27	81.32	85.23	74.58	87.12	77.35
SemiCD-VL + Ours	**89.90**	**81.95**	**90.35**	**82.88**	**87.69**	**77.92**	**89.05**	**80.14**
UniMatch	85.23	78.42	85.94	79.01	86.15	76.22	87.43	78.67
UniMatch + Ours	**87.93**	**79.95**	**88.12**	**80.23**	**88.31**	**78.85**	**89.17**	**81.03**

G. Efficiency Analysis

To evaluate the computational efficiency of our method, we conduct a comparative analysis on parameters, computational complexity (FLOPs), training time, and inference time of different methods on the LEVIR-CD dataset. Table V shows the performance of various methods on these key metrics. From the table, we can observe that our method has 40.47 M parameters and 29.10 G FLOPs, which is comparable to methods like s4GAN and UniMatch, but significantly lower than the 73.23 G FLOPs of SemiCD and RCL methods. In terms of training efficiency, our method requires 394.5 s/epoch, showing clear advantages compared to CutMix-CD's 745.95 s and SemiCD-VL's 971.88 s. For inference time, our method achieves 38.20 ms, demonstrating competitive performance among all comparison methods.

H. Generalization Evaluation

To validate the generalization capability of our uncertainty-guided strategy, we apply this strategy to different baseline semisupervised models and conduct comparative experiments on LEVIR-CD and WHU-CD datasets. Table VI presents the performance comparison results. The experimental results demonstrate that our uncertainty-guided strategy achieves consistent performance improvements across all tested baseline models. On the LEVIR-CD dataset, the FPA model shows significant improvement after applying our strategy, with $F1$ score increasing from 84.94% to 86.40% and IoU from 73.83% to 75.99%. SemiCD-VL and UniMatch(PSPNet), as strong baseline methods, also obtain substantial improvements. These results fully demonstrate that our proposed uncertainty-guided semisupervised learning strategy possesses good generalization and scalability.

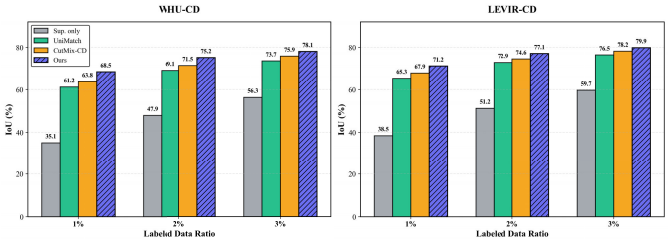


Fig. 9. Performance under extreme label scarcity (1%–3%).

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT UNCERTAINTY MEASURES ON LEVIR-CD DATASET. ALL METHODS USE THE SAME BASELINE FRAMEWORK WITH IDENTICAL HYPERPARAMETERS, ONLY THE UNCERTAINTY MEASURE DIFFERS

Uncertainty Measure	5% IoU	10% IoU	Formula
Entropy	80.23	81.45	$-\sum_c p_c \log p_c$
Max Probability	79.89	81.12	$1 - \max(p_c)$
Ada-Uncertainty Map	81.01	82.18	$e^{\alpha(1-\lvert p_0 - p_1\rvert)} \cdot H_{rb}$
Ours	**81.56**	**82.55**	$\mathbf{1 - \lvert p_0 - p_1 \rvert}$

Fig. 10 shows the qualitative comparison results between our method and different baseline approaches. It can be observed that baseline methods (FPA, SemiCD-VL, UniMatch) exhibit obvious detection errors in change boundaries and detailed regions, manifested as blurred boundaries and missed small targets. On the contrary, our USCD method can more accurately detect change regions, particularly showing significant improvements in building boundaries and complex change patterns.

The uncertainty maps reveal that baseline methods' high-uncertainty regions are mainly concentrated at incorrectly predicted boundary locations, while our method's uncertainty distribution is more reasonable, with high-uncertainty areas primarily appearing at genuinely difficult change boundaries. This validates that our uncertainty-guided mechanism can effectively identify and handle difficult regions, thereby improving overall detection performance.

I. Uncertainty Measure Comparison

To validate the effectiveness of our proposed uncertainty measure, we conduct controlled experiments where only the uncertainty quantification method varies while all other components remain identical. Table VII compares our measure against representative uncertainty quantification methods on the LEVIR-CD dataset.

We compare with three baseline measures: standard entropy, maximum probability-based uncertainty, and the recent Ada-Uncertainty Map [68]. For the Ada-Uncertainty Map, we follow the parameter settings from the original article. As

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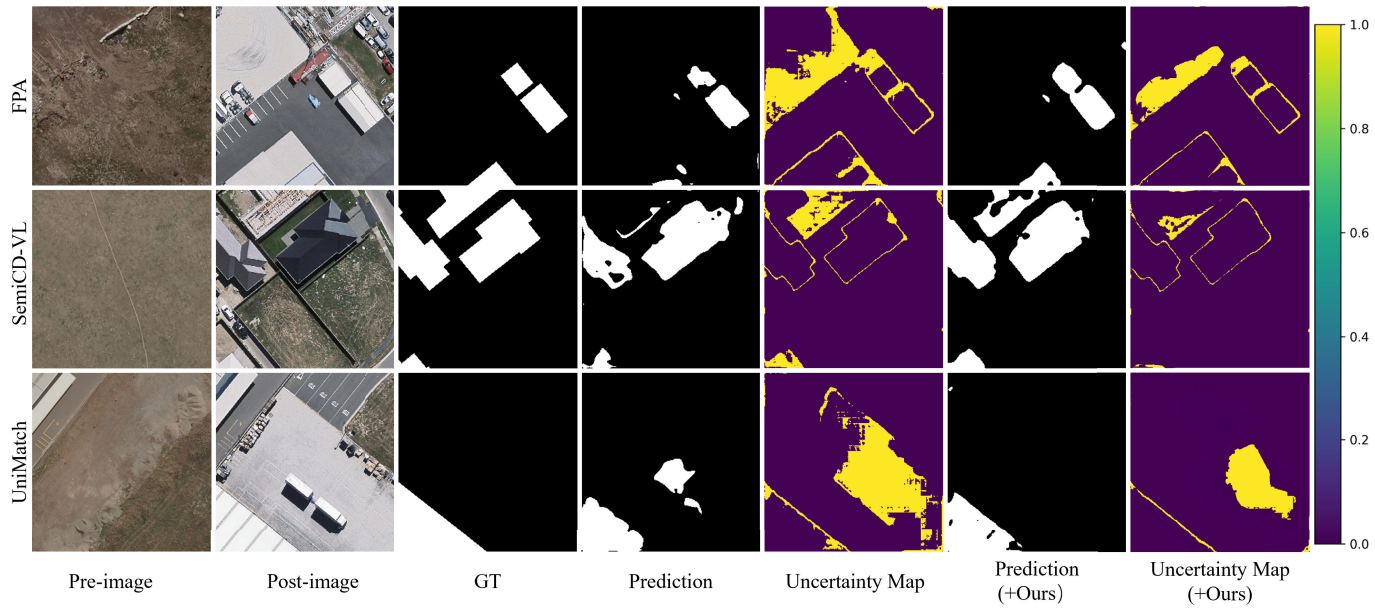


Fig. 10. Qualitative comparison results of different methods on WHU-CD.

TABLE VIII
UNCERTAINTY-DRIVEN VERSUS RANDOM SELECTION IN UAPA AND DRCL (UGLR KEPT INTACT IN ALL VARIANTS). RANDOM-BOTH: RANDOM SELECTION IN BOTH UAPA AND DRCL; RANDOM-UAPA: RANDOM SELECTION IN UAPA ONLY; RANDOM-DRCL: RANDOM SELECTION IN DRCL ONLY; OURS: UNCERTAINTY-DRIVEN SELECTION IN BOTH

Method	LEVIR-CD				WHU-CD			
	5%		10%		5%		10%	
	F1	IoU	F1	IoU	F1	IoU	F1	IoU
Random-Both	88.12	78.34	89.23	80.12	86.45	75.23	87.89	77.34
Random-UAPA	88.89	79.23	89.78	80.89	87.23	76.45	88.56	78.67
Random-DRCL	89.34	80.12	90.23	81.34	87.89	77.34	89.12	79.89
Ours	**90.22**	**81.56**	**90.99**	**82.55**	**89.41**	**81.52**	**90.97**	**83.01**

shown in the table, our simple formulation $U = 1 - |P_0 - P_1|$ consistently achieves the best performance across different annotation ratios, outperforming Ada-Uncertainty Map by 0.55% and 0.37% IoU at 5% and 10% labeled data, respectively.

The superiority of our measure over Ada-Uncertainty Map—which incorporates class rebalancing and exponential amplification—demonstrates that explicitly modeling the probability difference between changed and unchanged classes is inherently more effective for binary CD. Our measure directly captures the essential ambiguity at decision boundaries where $P_0 \approx P_1$, achieving maximum uncertainty (U = 1) when $P_0 = P_1 = 0.5$, without requiring additional parameters or computational overhead.

J. Validity of Uncertainty-Driven Selection

To isolate the actual contribution of the uncertainty signal in region selection, we replace the uncertainty-driven selection strategy in UAPA and DRCL with random selection while keeping UGLR intact in all variants. Three comparison groups are constructed: Random-Both replaces uncertainty-driven selection in both UAPA and DRCL with random selection; Random-UAPA applies random selection only in UAPA while retaining uncertainty-driven mining in DRCL; Random-DRCL applies random selection only in DRCL while retaining uncertainty-driven protection in UAPA. As shown in Table VIII, our full uncertainty-driven method consistently outperforms all random counterparts across both datasets and annotation ratios. On LEVIR-CD with 5% labeled data, Random-Both achieves 88.12% $F1$, which is 2.10% below our full model (90.22%), demonstrating that random selection fails to effectively identify spatially heterogeneous difficult regions even when UGLR is present. Random-UAPA (88.89% $F1$) and Random-DRCL (89.34% $F1$) show intermediate performance, confirming that uncertainty-driven selection is indispensable in both modules independently. Notably, the larger gap between Random-UAPA and Ours (+1.33% $F1$) compared to Random-DRCL and Ours (+0.88% $F1$) suggests that UAPA benefits more from uncertainty guidance, as protecting genuinely difficult regions during augmentation is crucial for preserving discriminative information. Consistent trends are observed on WHU-CD, where Random-Both lags behind Ours by 2.96% $F1$ at 5% annotation. These results collectively validate that the performance improvements of USCD are primarily driven by the uncertainty signal itself rather than the selection mechanism, confirming the necessity of uncertainty-guided selection as the core design principle.

K. Effect of Window Number N

Table IX reports the performance of UAPA under different window numbers N on three datasets with different change ratios (CR). The results consistently validate our CR-guided selection strategy: for the sparse-change dataset S2Looking

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TABLE IX
ABLATION STUDY ON WINDOW NUMBER N ACROSS DATASETS WITH DIFFERENT CR. ALL RESULTS ARE REPORTED WITH 5% LABELED DATA

N	S2Looking (CR=1.21%)		SYSU-CD (CR=21.34%)		LEVIR-CD (CR=4.59%)		Epoch (s) (LEVIR-CD)
	F1	IoU	F1	IoU	F1	IoU	
4	55.51	39.02	**79.94**	**66.83**	90.01	81.33	393.34
8	55.82	39.18	79.80	66.50	**90.22**	**81.56**	394.53
16	**55.99**	**39.39**	79.57	66.39	90.08	81.40	395.92

(CR = 1.21%), a finer partition of $N = 16$ yields the best performance, as smaller windows preserve the weak uncertainty signals of isolated change pixels that would otherwise be diluted in coarser windows. For the moderate-change dataset LEVIR-CD (CR = 4.59%), $N = 8$ achieves the best tradeoff between region precision and signal reliability. For the dense-change dataset SYSU-CD (CR = 21.34%), a coarser partition of $N = 4$ performs well, since fine-grained windows overfragment large contiguous change regions and reduce the discriminability of the uncertainty scores. It is worth noting that the computational overhead introduced by the window partition varies negligibly across different N values, since the dominant cost lies in the network forward pass rather than the window aggregation and sorting operations.

V. CONCLUSION

In this article, we propose a novel semisupervised CD framework, USCD, which precisely identifies and reinforces learning in difficult regions through uncertainty guidance mechanisms. The proposed method integrates UAPA, DRCL, and UGLR into a unified framework, effectively alleviating the problem of insufficient learning in difficult regions in semisupervised CD. The method achieves competitive performance on six benchmark datasets, demonstrating strong effectiveness under annotation-scarce scenarios, with especially pronounced improvements in challenging regions such as change boundaries and subtle changes. Cross-domain experiments and method generalization validation further prove the stability and scalability of the framework. Despite these promising results, the square-window design in UAPA may not align well with the irregular contours of real change objects and may introduce background contamination. Future work will focus on object-aware adaptive augmentation, where pseudolabels or boundary cues are used to guide copy-paste regions along object contours. We will also extend USCD to semantic CD by designing uncertainty measures and contrastive learning strategies for multiclass prediction distributions.

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